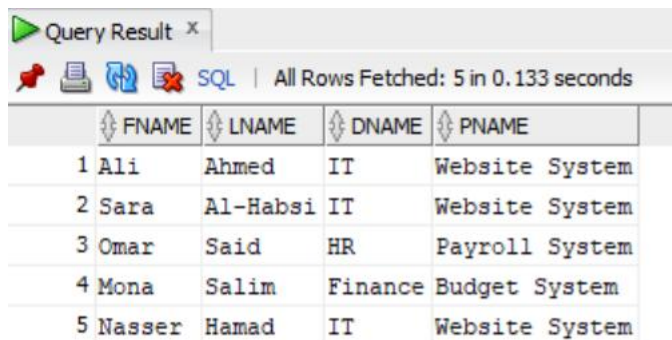


SQL JOIN Implementation

1- INNER JOIN

```
SELECT E.FNAME, E.LNAME, D.DNAME, P.PNAME FROM EMPLOYEE E  
JOIN DEPARTMENT D ON E.DNUM = D.DNUM JOIN PROJECT P ON D.DNUM =  
P.DNUM;
```



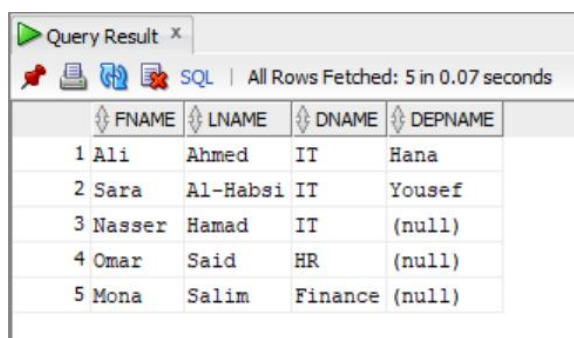
	FNAME	LNAME	DNAME	PNAME
1	Ali	Ahmed	IT	Website System
2	Sara	Al-Habsi	IT	Website System
3	Omar	Said	HR	Payroll System
4	Mona	Salim	Finance	Budget System
5	Nasser	Hamad	IT	Website System

Screenshot Caption:

This query uses INNER JOIN to display employees, their departments, and projects. Only matching records from all tables are returned.

2- LEFT JOIN

```
SELECT E.FNAME, E.LNAME, D.DNAME, DP.DEpname  
FROM EMPLOYEE E LEFT JOIN DEPARTMENT D ON E.DNUM = D.DNUM  
LEFT JOIN DEPENDENT DP ON E.SSN = DP.SSN;
```



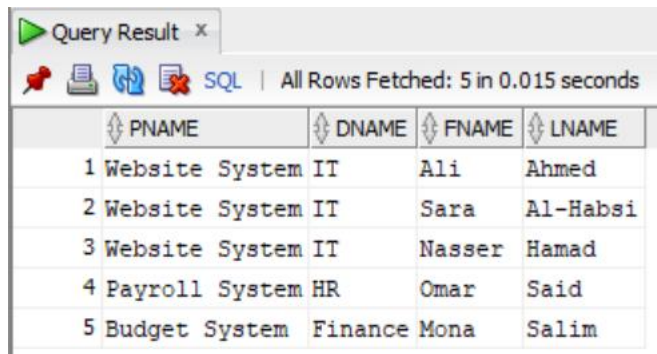
	FNAME	LNAME	DNAME	DEPNAME
1	Ali	Ahmed	IT	Hana
2	Sara	Al-Habsi	IT	Yousef
3	Nasser	Hamad	IT	(null)
4	Omar	Said	HR	(null)
5	Mona	Salim	Finance	(null)

Screenshot Caption:

This query uses LEFT JOIN to display all employees with their department and dependent information, including employees who do not have dependents.

3- RIGHT JOIN

```
SELECT P.PNAME, D.DNAME, E.FNAME, E.LNAME FROM EMPLOYEE E RIGHT  
JOIN DEPARTMENT D ON E.DNUM = D.DNUM RIGHT JOIN PROJECT P  
ON D.DNUM = P.DNUM;
```



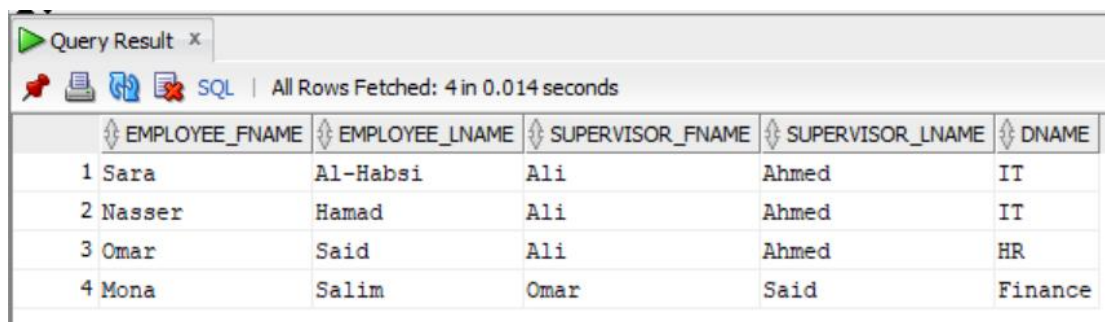
	PNAME	DNAME	FNAME	LNAME
1	Website System	IT	Ali	Ahmed
2	Website System	IT	Sara	Al-Habsi
3	Website System	IT	Nasser	Hamad
4	Payroll System	HR	Omar	Said
5	Budget System	Finance	Mona	Salim

Screenshot Caption:

This query uses RIGHT JOIN to display all projects with their departments and employees, including projects that have no assigned employees.

4- SELF JOIN

```
SELECT E.FNAME AS EMPLOYEE_FNAME, E.LNAME AS EMPLOYEE_LNAME,  
S.FNAME AS SUPERVISOR_FNAME, S.LNAME AS SUPERVISOR_LNAME,  
D.DNAME FROM EMPLOYEE E JOIN EMPLOYEE S ON E.SUPER_SSN = S.SSN  
JOIN DEPARTMENT D ON E.DNUM = D.DNUM;
```



	EMPLOYEE_FNAME	EMPLOYEE_LNAME	SUPERVISOR_FNAME	SUPERVISOR_LNAME	DNAME
1	Sara	Al-Habsi	Ali	Ahmed	IT
2	Nasser	Hamad	Ali	Ahmed	IT
3	Omar	Said	Ali	Ahmed	HR
4	Mona	Salim	Omar	Said	Finance

Screenshot Caption:

This query uses SELF JOIN to display each employee with their supervisor and department information using the Employee table twice.